MATHEMALCHEMY

A SCAVENGER HUNT

Mathemalchemy

Mathemalchemy is the work of 24 artists and mathematicians. **Start at mathemalchemy.org** — the creators' own site, and the fullest account of the installation, with photographs, videos and fabrication notes for every scene. Companion pages, with activities and classroom connections, are at csvaughen.github.io/mathemalchemy-docent-guide.

A dozen creatures and curiosities are hiding in the exhibit, each with a piece of mathematics behind it. Solve each riddle, find the thing it describes, and write down what you found. **Find as many as you can** — and look, please don't touch.

1 I never rush, yet always advance — halfway, then halfway, I'll get there… perchance. Seven-sided tiles on my shell, and a paradox walk — find the slowpoke who out-thinks the clock.

I FOUND _____

2 A baker with whiskers rolls out the dough, stamping small π's in a tidy tight row — no crumb left over, no gap you can see. Name the chef who serves chaos for tea.

I FOUND _____

3 Zoom in forever and I look the same — a shoreline of copies, a self-similar frame. My portrait's the doorknob that welcomes you in. Look close: what fractal is waiting within?

I FOUND _____

4 The herons haul nets from the bay, dripping wet, tangled with loops that no sailor'd forget. They're not catching supper — the truth's stranger yet. Find the birds and see what's really in the net.

I FOUND _____

5 These small striped sorters ignore 4 and 9; they only keep numbers that won't split fine — divisible by themselves and by one alone. Which critters keep count where the garden has grown?

I FOUND _____

6 It's always midwinter upon this odd hill, where snowflakes have edges that crinkle and fill — endless in border, yet finite in space. Spot one of the flakes in this frostbitten place.

I FOUND _____

7 A hotel whose rooms never once say 'no vacancy' — even jam-packed, there's room for infinity. Find the sign where the beds and the rooms never end.

I FOUND _____

8 On a string in the breeze the tortoise lets fly a triangle built of triangles on high — holes within holes, the same shape all the way. Look up — find the kite that's been flying all day.

I FOUND _____

9 An amateur's pentagons, stitched with great pride, tile a whole floor with no gaps left to hide. Peer in and look down: in which room is that floor, where five sides meet five, and then five sides once more?

I FOUND _____

10 A little shop stocked with the strangest of wares — mathematical oddities, curios, rares. Named for a mind that loved puzzles and play: find the shop, and write one odd thing on display.

I FOUND _____

11 The baker's small helper, with whiskers and all, stamps treats that will tessellate, wall to wall. Two mathematicians hide in this pair of names — spot the wee helper who shares in the games.

I FOUND _____

12 Tall by the shore with a beam sweeping wide, a spiraling ramp winding up the outside. What beacon stands watch where the land meets the tide?

I FOUND _____

the mathematics is overleaf — hunt first →

The mathematics behind the hunt

Turn here *after* you have hunted — each note names the mathematics the object is really about, and each one goes further than this page does.

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1 Tess the Tortoise illustrates Zeno's dichotomy paradox — to arrive you must first cross half the distance, then half of what is left, then half again, forever; the pavers of her path shrink to match. The seven-sided tiles on her shell are a heptagonal tiling of the hyperbolic plane, drawn in the Poincaré disk model. Three regular heptagons meeting at a corner come to 3 × 128.6° = 385.7°, which overshoots the 360° a flat surface has to offer — so they cannot tile a plane, but on a negatively curved surface they close perfectly.

2 The baker is Arnold, named for Vladimir Arnold (1937–2010), who worked on the mathematics of motion — planets, fluids, anything that changes over time. A baker mixes dough by stretching and folding it back on itself, and that is the heart of how chaos mixes things. His famous picture does it to a cat: stretch and slant the drawing across a square, slide the overhang back in, repeat, and in a few rounds the cat is an even gray. That is Arnold's cat map, and several of its stages ring the feet of the adult silhouette who releases the cavalcade of pages. But the picture has finitely many pixels, and stretching only moves them — it never loses one. There are only so many arrangements to pass through, so keep iterating and the cat must come back, whole. You found Arnold the cat, now go find the drawing of Arnold's cat map.

3 The Mandelbrot set, a fractal that holds scaled copies of itself at every magnification. Its border is infinitely long and infinitely detailed.

4 Knots — knot theory is the study of which tangles can be undone without cutting the rope, and how to tell two knots apart when they look nothing alike.

5 Chipmunks sorting prime numbers. Their neighbors in the garden are working the same problem a different way, at a Sieve of Eratosthenes — worth finding both.

6 The Koch snowflake: a curve of infinite perimeter enclosing a finite area. You could never walk its edge, but you could paint its inside.

7 Hilbert's Hotel, David Hilbert's illustration of countable infinity. A hotel with infinitely many rooms, every one of them occupied, can still take a new guest: move room 1's guest to room 2, room 2's to room 3, and so on forever. Nobody is turned out, and room 1 is free. Infinite collections do not behave like finite ones.

8 The Sierpiński tetrahedron — a tetrahedron built from four half-size copies of itself, each built from four more. Halving an edge quarters a face's area, and there are four copies, so the total surface area never changes from one stage to the next; but halving an edge cuts each piece's volume to an eighth, and four eighths is a half, so the volume falls to zero. All surface and no weight — exactly what a kite wants.

9 Marjorie Rice's pentagon tilings. A San Diego homemaker with no mathematics past high school found four new ways for a convex pentagon to tile the plane, working at her kitchen table in a notation she invented for herself, after the professionals had stopped looking. Fifteen types are now known.

10 Conway's Curios, named for John Horton Conway, who invented the Game of Life and the surreal numbers and treated mathematics as a source of toys. What sits on the shelves changes as the exhibit travels: Klein bottles, a Möbius band, an Alexander horned sphere, Borromean rings, polyhedra of several families.

11 Mo-u-se's odd spelling hides a second mathematician, Jürgen Moser (1928–1999). Put the assistant beside the baker and you have two of the three names on the Kolmogorov–Arnold–Moser theorem — the K is Andrei Kolmogorov, Arnold's teacher, who had the original idea; Arnold and Moser carried it through. It is part of why a solar system can stay stable for an extremely long time even though every planet keeps tugging on every other one.

12 The Lighthouse — a regular heptagon forms the base, the smallest regular polygon that cannot be drawn with straightedge and compass alone. The body is a *hyperboloid*, perturbed. The ramp is welded onto the outside, but the route it takes is not arbitrary: it follows a **line of principal curvature**. At any point a curved surface bends most sharply in one direction and most gently in another, and a curvature line always runs along one of those two. Follow one and a flat steel strip need only bend, never twist — which is how a nine-foot tower gets welded together out of flat stock. At the top, a Fresnel lens and a dodecahedron house the light, and a carved sphere throws its pattern onto the ceiling by stereographic projection: sizes come out distorted, but every angle is exactly right. That is the same property that makes the Mercator map good for steering a ship. Look at the dodecahedron's two colours of glass: the line dividing them is a **geodesic** — dead straight, in the sense that if you unfolded the faces flat as you crossed each edge it would be a ruler line. Set off that way from a corner of a cube and you always run into another corner. The dodecahedron is the only Platonic solid on which you can leave a corner and return to it having missed every other one, and nobody proved that until 2020.